

1) explain about interface sample program. program should have a class with interface implementation

• An interface in Java is a blueprint of the class. It is a abstract methods. A class implements a interface by providing the implementation for it methods.
It used to multiple inheritances it used abstract it used to code Reusability.

Syntax

```
interface name {  
    void method ( ) ;  
}
```

```
class classname implements interface name {  
    public void method name ( ) {  
    }
```

Program: Package basic;

```
interface Animal {  
    void sound ( ) ;
```

```
}  
class cat implements Animal {  
    public void sound ( ) {
```

```
        System.out.println ("dog bow bow");
```

```
    public static void main ( ) args {
```

```
        Dog D = new Dog ( );
```

```
        D.sound ( ) ;  
    }
```


output:

Dog bow bow

2) what is the difference b/w method overloading and method overriding? Explain the same with programs.

overloading

- 1) Compile-time Polymorphism
- 2) inheritance not required
- 3) It is in the same class
- 4) If parent class it is in parent class

overriding

- 1) Runtime Polymorphism
- 2) inheritance not required
- 3) It happens in parent not in child class
- 4) It child class means it is in parent class not in child class

overloading
package Basic;
class Add {

void add(int a, int b)
System.out.println("sum = a + b");

}

void add(int a, int b, int c)
System.out.println("sum = a + b + c");

}
public class overloading {
public static void main (String[] args)
System.out.print("sum")
object = new Add;

object add (10, 20);
object add (10, 20, 30)

3

output

sum = 30

sum = 60

3) Difference b/w System error and Exception Handling
Arithmetic and

overriding:

class Add {

void add (int a, int b)

System.out.println ("Sum = a+b");

}

void add (int a, int b, int c);

System.out.println ("Sum = a+b+c");

3

void add public class overriding {

public static void main (String [] arg) {

public static System.out.print ("Sum")

Add object = new Add; objectadd = (10, 20)

Add object = new Add; objectadd = (10, 20, 30)

}

}

output : sum = 30
sum = 60

3. Difference between system error and exception handling
Arithmetic and Array Index out of bounds exception in a
single program.

1) occurred in the Java program incorrect syntax in the program.

2) missing semicolon, close bracket, etc.

3) The before the run (or) execution the error will occur.

1) occurred in a program execution

2) when run the program the execution after the error will occur

3) the error are try catch error, and throw, throws, finally - finally.

Program:

```
Package basics;  
Public class Exception;  
Public void static void (String [] args);  
// Arithmetic exception.
```

```
// int a = 10  
// int b = 0  
int c = a/b
```

```
System.out.print(c);
```

```
}
```

```
System.out.println("Arithmetic exception is catch");  
Public class Exception;
```

```
Public static void (String [] args);  
int arr [] = {10, 20, 30};
```

```
System.out.println(arr[5]);
```

```
System.out.println("Array Index out of bounds");
```

output:
Arithmetic exception caught
Array Index out of Bounds caught